Vulnerability Assessment Report

1st January 20XX

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. NIST SP 800-30 Rev. 1 is used to guide the risk analysis of the information system.

Purpose

A vulnerability assessment was due for this company since a remote database server is heavily relied on and since it is open to the public it is a serious vulnerability. Potential customers are found using this database and if it is compromised it can result in a catastrophic halt on business operations and assets.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor	*Obtain sensitive information via exfiltration*	*3*	*3*	*9*
Storage	*Server runs out of storage*	*2*	*2*	*4*
Power Outage	*Power running to the server short circuits*	*1*	*3*	*3*

Approach

After analyzing the possible threats and their risks, a couple stand out. A competitor can access our customer information since it is public. This is very likely since any competitor would like to have the upper hand against us and take our customers. The likelihood and severity of this is very high. Since we have a server with only 128GB of memory our storage would fill up fast and we would not be able to store more information. This wouldn't affect the information we already have but would not let us gather more. Finally a power outage would stop business operations even though it is more on the unlikely side. Our server is the only way to access our data since we don't have a backup.

Remediation Strategy

To remedy our highest risk threat in our current situation we must use a combination of principle of least privilege and MFA. If our servers are removed from public access and a multi factor authentication system is put into place, then our competitors won't be able to access our customer data. Using the principle of least privilege, all employees will be given the minimal access and authorization required to complete their tasks. Using MFA will guarantee that our employees are authenticated and getting the access they require.